ROLLUP

⭐⭐⭐⭐ Advanced Reporting Feature

ROLLUP is used when you want SQL to generate subtotals and a grand total automatically, without writing multiple queries.

Think of reports like:

- Sales by City → State → Country
- Revenue by Product → Category
- Employees by Team → Department
- Orders by Month → Year

Instead of manually calculating totals, ROLLUP does it for you.

14.17.1 Why ROLLUP Exists

Suppose we have a sales table.

Country	State	City	Sales
India	Delhi	New Delhi	100
India	Delhi	Dwarka	200
India	Bihar	Arrah	150
India	Bihar	Patna	250

Normal GROUP BY

```
SELECT state,
       city,
       SUM(sales)
FROM Sales
GROUP BY state, city;
```

Output

State	City	Sales
Delhi	New Delhi	100
Delhi	Dwarka	200
Bihar	Arrah	150
Bihar	Patna	250

Useful.
But business reports usually also need

- Delhi Total
- Bihar Total
- India Total

Without ROLLUP, we'd need several extra queries.

14.17.2 Syntax

```
SELECT columns,
       aggregate
FROM table
GROUP BY ROLLUP(column1,column2,...);
```

14.17.3 Two-Level ROLLUP

```
SELECT state,
       city,
       SUM(sales)
FROM Sales
GROUP BY ROLLUP(state,city);
```

Output

State	City	Sales
Bihar	Arrah	150
Bihar	Patna	250
Bihar	NULL	400
Delhi	Dwarka	200
Delhi	New Delhi	100
Delhi	NULL	300
NULL	NULL	700

> Notice the NULL rows.
> They are generated by SQL.

14.17.4 Understanding NULL Rows

```
Bihar
    Arrah        150
    Patna        250
----------------------
Bihar Total      400

Delhi
    Dwarka       200
    New Delhi    100
----------------------
Delhi Total      300

----------------------
Grand Total      700
```

The generated rows appear as

`City = NULL`

because there is no specific city.

ROLLUP follows hierarchy from left to right.

`ROLLUP(state, city)` means

State
↓
City

SQL computes

State + City
↓
State Total
↓
Grand Total

What is ROLLUP?

Generates hierarchical subtotals and a grand total.

Why does ROLLUP produce NULL rows?

NULL indicates a subtotal or total level where that grouping column no longer has a single value.

Is ROLLUP hierarchical?

Yes.
Left-to-right hierarchy.

Does column order matter?

Absolutely.

`ROLLUP(state,city)` is different from `ROLLUP(city,state)`

Why CUBE Exists

Suppose

Department	City	Salary
ECE	Delhi	70
ECE	Mumbai	60
CSE	Delhi	90
CSE	Mumbai	80

Business may ask

- Salary by department and city
- Department totals
- City totals
- Overall total

> Without CUBE
> Need many separate queries.

14.18.2 Syntax

```
SELECT columns,
       aggregate
FROM table
GROUP BY CUBE(column1,column2);
```

14.18.3 GROUP BY Only

```
GROUP BY department,city
```

Produces

Department	City
ECE	Delhi
ECE	Mumbai
CSE	Delhi
CSE	Mumbai

Only detailed groups.

14.18.4 ROLLUP

```
ROLLUP(department,city)
```

Produces

Department
↓
City
↓
Department Total
↓
Grand Total

Output

Department	City
ECE	Delhi
ECE	Mumbai
ECE	NULL
CSE	Delhi
CSE	Mumbai
CSE	NULL
NULL	NULL

> Notice
> No city totals.

14.18.5 CUBE

```
GROUP BY CUBE(department,city);
```

Output

Department	City
ECE	Delhi
ECE	Mumbai
CSE	Delhi
CSE	Mumbai
ECE	NULL
CSE	NULL
NULL	NULL
NULL	Delhi
NULL	Mumbai

> Notice the extra rows
>
> NULL
> Delhi
>
> These are
> City Totals.

ROLLUP	CUBE
Department + City ↓ Department Total ↓ Grand Total	Department + City ↓ Department Total ↓ City Total ↓ Grand Total

Number of Grouping Sets

For n columns:

CUBE
↓
2^n grouping sets

Examples

Columns	Grouping Sets
1	2
2	4
3	8
4	16

This exponential growth is why large CUBEs can become expensive.

14.18.9 Performance

CUBE generates far more rows than ROLLUP.

Example
2 columns

ROLLUP	**CUBE**
↓	↓
3 grouping levels	4 grouping sets

With more columns, the difference grows rapidly.